WEEK 6 · DAY 3 · THE LAST SESSION

Demo Day + Project Clinic

Six weeks. One tool for your unit. Five minutes to show it.

NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy



TODAY'S PLAN · 2 HOURS

Fix it. Show it. Celebrate it.



⏱️ 10 min welcome · 🩺 30 min clinic · 🎤 60 min demos · 🎉 20 min wrap-up

TWO PEOPLE YOU ALREADY KNOW

The MC and the uncle 🎉



The owambe MC

"Groom's friends, **five minutes**! After that, the DJ takes the floor." And he means it: the music starts on time.

keeps to time



The uncle in the front row

"Ehen, my son. Tell us in **one sentence**: what does it do, and why should we care?"

wants the point

Today you satisfy both: **five minutes**, and a point your unit head can repeat.

LOOK HOW FAR YOU'VE COME

18 days in one picture 🛣️



WEEK 1

What does this series look like?

- Colab + first series
- Charts that don't lie
- Rolling windows



WEEK 2

What is it made of?

- Trend · season · noise
- Autocorrelation
- Quiz 1 + your brief



WEEK 3

When is a reading wrong?

- Alarms that cry wolf
- Control charts
- Drift + CUSUM



WEEK 4

Can a machine spot the odd one?

- Scoring + Quiz 2
- Isolation forest
- Feature engineering



WEEK 5

Is the test honest? What comes next?

- No expo + Quiz 3
- Baselines
- Regression + ranges



WEEK 6

When will it break? Show us!

- Backtesting
- RUL + Quiz 4
- **Demo day 🎤**



Every project uses pieces from **at least three** of these weeks.

THE FINAL PROJECT, STAGE BY STAGE

Four checkpoints done. Today is the fifth.

🎯
Stage 0
Choose
brief + unit head's OK
✅ Week 2 · Day 3

📥
Stage 1
Understand
load, profile, plot
✅ Week 3 · Day 3

🛠️
Stage 2
Build
your method
✅ Week 4 · Day 3

⚖️
Stage 3
Test honestly
split by time + baseline
✅ Week 5 · Day 3

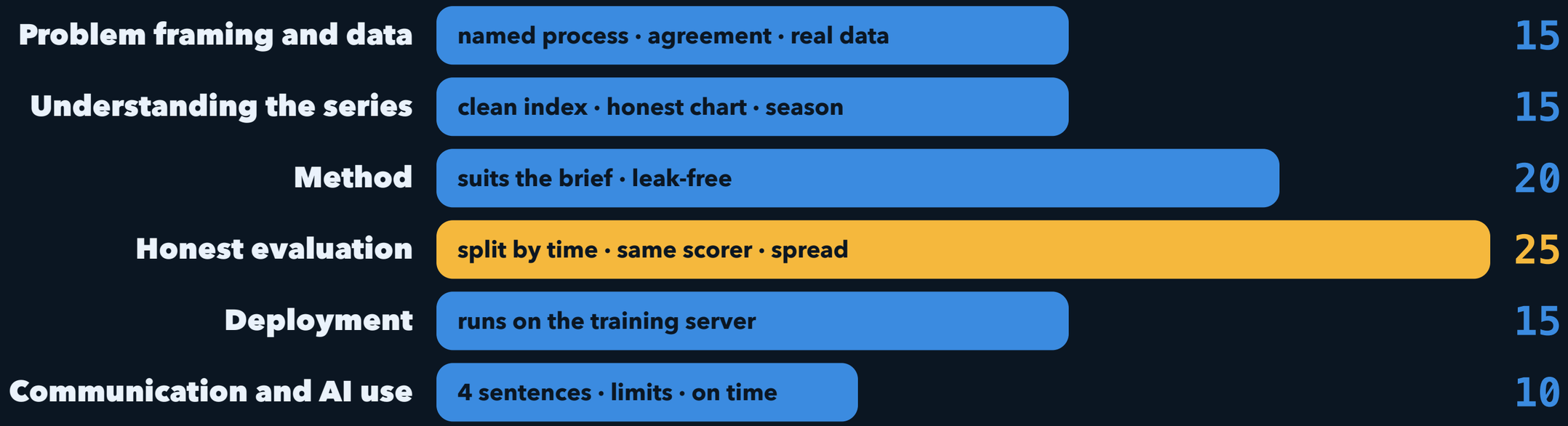
🚀
Stage 4
Deploy
on the training server
by today

🎤
Stage 5
Present
report + 5-minute demo
🎤 TODAY

Today you bring: a **working tool**, the **2-4 page report** and a **5-minute demo**.

HOW IT'S GRADED · 100 POINTS

The biggest slice is honesty ⚖️



No model has to win. **The test has to be honest** (25 of 100 points).

READ THIS TWICE

What costs marks, and what doesn't



Automatic penalties

a random train/test split
a feature that uses the future
an accuracy figure for rare events
a remaining-life date with no window
real data committed to GitHub



Not penalised

a model that **loses** to the simple baseline,
as long as the comparison is fair
and you **recommend the simpler method**.

We did exactly that in W5D1 and W6D1.

PART 1 · 30 MINUTES

The clinic 

Last fixes before the stage. Go where you need help most.

FOUR HELP STATIONS · PICK THE ONE YOU NEED

Clinic stations



1

Data & leaks

hygiene report · leak_test() on every feature · the time zone written down



2

Honest test

split assert · same period, same scorer · spread or coverage reported



3

Deployment

runs on the training server · README · cron or Streamlit · no raw data in Git



4

The story

4 sentences · 3 limitations · a timed 5-minute run-through

15 minutes at a station, then **rotate** or rehearse with a partner. Bring your laptop and your report.

LET THE COMPUTER BE YOUR REVIEWER

The self-check notebook 👩🏾‍⚕️

`hygiene()`
index · order · repeats · holes · frequency · time zone

`leak_test()`
hide the future, rebuild the feature: did it change?

`check_temporal_split()`
train before test, with the embargo, or it fails loudly

`compare()`
every method, same weeks, same scorer, best first

`coverage()`
did the 90% band catch about 90% of held-out weeks?

`recommendation()`
4 sentences, one number, one action

Rehearse on Colab with course data. Then **download it** and run it on the training server with your real data.

THE LEAK TEST, DONE BY THE COMPUTER

Hide the future. Did the feature change? 🙈



For several weeks **t**: blank out week t's answer and every row after it, rebuild the feature, compare. If the value moved, the feature **needed something you couldn't know**.

`last week: shift(1)`	✅ **honest**
`same week last year: shift(52)`	✅ **honest**
`terminals this week`	✅ **honest**
`next week: shift(−1)`	🚨 **leaks**
`4-week mean, centred`	🚨 **leaks**
`4-week mean, including this week`	🚨 **leaks**

REHEARSAL · ADA'S DRAFT PROJECT

What the self-check found 🔎

👯🏾‍♀️

1

week exported twice

🕳️

1

week that never arrived



shuffled

training reaches 22 Dec 2025, the test starts 10 Jun 2024



6/8

weeks inside her “90%” band (±3)

Fixed: fill the gap, split by time, band from **honest past misses** (±7.0): 8/8.

PART 2 · 60 MINUTES

Demo time 🎤

5 minutes to show it. 1 minute for one question. The timer is the MC.

THE 5-MINUTE DEMO, SECOND BY SECOND

Six beats. No slide is longer than its beat. ⏱️

0:00	0:45	1:30	2:15	3:15	4:15
Problem	**Data**	**Method**	**Honest test**	**The tool, live**	**Recommend**
unit · named process · who uses it 45 s	source · span · cadence · what you fixed 45 s	what, and why it suits the brief 45 s	split by time · baseline · the table 60 s	dashboard or today's report 60 s	4 sentences + one limitation 45 s

5:00 ⏰

🧪 **Show the tool live**, not a screenshot. Have the screenshot as backup.

📊 **One table** for the honest test: your method vs the baseline, same period.

🗣️ **End on the recommendation**, not on “any questions?”

The 4-sentence recommendation 🗣️

1	**Situation**	The NOC sets its month-ahead duty roster from last week's ticket count.
2	**Observation**	Backtested from 45 starting weeks, last year's tickets per terminal times this week's network missed by 3.2 tickets a week, 8 weeks ahead, against 6.2 for “same as last week”.
3	**Implication**	A roster built on the rate forecast would be about half as far off, at no extra cost.
4	**Recommendation**	Use the rate forecast for the monthly roster, keep “same as last week” for next week, and re-run the backtest every quarter.

TWO MORE OBSERVATIONS FROM THIS COURSE

One number, said plainly 🔢



Battery (W6D2)

"Daily capacity first dipped below 80% on **14 May 2025** and was last at or above it on **24 Jul 2025**: a 72-day crossing window. A straight line on three years of data called **1 May 2025**, 13 days before the first dip."



Detector (W5D1)

"On the held-out months both detectors caught **2 of 2** events. The control chart raised **1** false alarm; the isolation forest raised **25**. We recommend the chart."

A window, not a single day. The same events, the same scorer. **Then** the verdict.

THE JUDGES HAVE SEEN THESE BEFORE

A strong demo vs the six traps



Strong

names the process and its owner
says what was fixed in the data
split by time, with the assert
beats (or honestly loses to) a baseline
a window or a band, with its coverage
runs live on the training server

🚨 Traps

- 🎲 **A random split:** shuffled weeks or hours: the score is expo
- 🙈 **No baseline:** “my model scores 4.0”… compared with what?
- 📏 **A band never checked:** “90%” with no coverage on held-out data
- 🎯 **Accuracy for rare events:** 99% by never raising an alarm: use event-level recall
- ☁️ **Real data in Colab or AI:** real NIGCOMSAT data stays on NIGCOMSAT systems
- 💻 **Laptop-only tool:** if it doesn't run on the server, it isn't deployed

SAY IT BEFORE THE UNCLE ASKS

Three honest limitations 🧭



What will it miss?

Sudden changes the calendar can't know about, such as a new region going live or a long outage.



How late / how far?

It looks 1 to 8 weeks ahead; its 90% band is about ±7 tickets a week.



What did you call normal?

It assumes the 2024–2025 rainy-season pattern repeats and that the terminal count is reported on time.

Unit heads trust the engineer who says **where the tool stops working**.

WHILE OTHERS PRESENT

Two stars and a wish ⭐⭐🙏

⭐

Star 1

one thing that was clear or clever

⭐

Star 2

one thing you'll steal for your own work

🙏

A wish

one thing that would make it stronger

Use the **peer scoring sheet**: the six rubric lines, a quick score, then your stars and wish.

Kind, specific, about the work. The instructor's mark is the official one.



Oya, you don finish!

Certificates · a group photo · a round of applause for every presenter.

🏅 certificates 📸 group photo 👏🏾 applause

THE COURSE ENDS. THE TOOL DOESN'T.

What next? 🚀

🔁

Keep it running

re-run the backtest or the remaining-life call every month or quarter

📨

Hand it over

send your unit head the report, the README and the link

📚

Keep the toolkit

rule books, the master glossary, every notebook



Ask for better data

time sync, longer archives, structured tickets (requirements PDF, section 8)

And remember: the **honest** result is always the respectable one.

SIX WEEKS IN THREE LINES

What you can now do



Read a series: clean it, plot it honestly, take it apart.



Catch what goes wrong, and score it with event-level rules.



Say what comes next, with a range, tested on the past first.

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